NETRA - Offline AI Vision Assistant

QUALCOMM (R) SNAPDRAGON (R) AI LAB BUILD & PRESENT CHALLENGE 2026

One-Line Pitch: Netra is an accessible, 100% offline AI vision assistant designed for low-vision, blind, and elderly users, converting live camera feeds into real-time spoken descriptions via on-device object detection and text recognition.

1. Problem Statement

Over 2.2 billion people worldwide live with vision impairment. In daily environments, low-vision and elderly individuals face constant accessibility friction:

- **Reading Printed Text:** Difficulty identifying medication labels, warning signs, nutrition panels, and printed documents.
- **Object Identification:** Challenges recognizing everyday physical objects, personal belongings, or obstacle hazards in immediate proximity.
- **Cloud & Network Dependence:** Traditional vision assistance tools rely on cloud APIs, creating latency, subscription costs, and severe privacy risks for personal camera feeds.
- **Complex Interfaces:** Mobile visual assistant tools often require multi-step navigation unsuitable for users requiring immediate hands-free audio feedback.

2. Proposed Solution

Netra addresses this challenge by providing a lightweight, completely local AI vision assistant running directly on Windows PCs. Netra captures video frames from a standard webcam, processes the image through an integrated on-device AI pipeline, and immediately speaks out a structured natural-language description.

Camera Feed	>	Object Detection	>	Text OCR	>	Local Synthesis	>	Speech Output
Webcam Input		YOLOv8-N ONNX		EasyOCR Engine		Response Composer		pyttsx3 / SAPI

3. Key Implemented Features

- **Real-time Object Detection:** Identifies physical objects using an optimized YOLOv8-N ONNX model (detector.onnx, 12.26 MB), outputting bounding box coordinates and COCO class labels.
- **Offline Text Recognition (OCR):** Integrates EasyOCR to recognize printed text from product packaging, signage, and documents without internet connectivity.
- **Deterministic Local Response Composer:** Synthesizes detected visual labels and OCR text into concise, natural spoken descriptions with zero LLM hallucination risk.
- **Offline Text-to-Speech (TTS):** Utilizes Windows SAPI via pyttsx3 for instant, clear audio playback.
- **High-Contrast Visual Dashboard:** Displays bounding boxes, real-time status overlay, detected items, extracted text, and active execution engine information on a dark high-contrast dashboard.
- **Single-Key Interaction:** Designed for extreme simplicity - press SPACE to look & speak, press Q/ESC to quit.

4. Technology Stack & Specs

Component	Technology / Model	Execution Mode & Status
Core Runtime	Python 3.10 / OpenCV 5.0	Verified local execution & camera capture
Inference Engine	ONNX Runtime 1.23	QNN EP pre-configured; CPU Fallback active
Object Detector	YOLOv8-N ONNX (12.26 MB)	Empirical ~40 ms latency (~25 FPS on CPU)
OCR Engine	EasyOCR (CPU Mode)	Offline text extraction (~1.5s latency)
Response Synthesis	Local Response Composer	Deterministic rule-based context merger
Speech Synthesis	pyttsx3 / Windows SAPI	Offline native Windows TTS audio output

5. Qualcomm Snapdragon (R) Hardware Relevance

Netra's core inference engine (app/engine.py) is structured specifically for Qualcomm Snapdragon ARM64 Copilot+ PCs running Windows 11. The model loader actively requests Qualcomm's QNNExecutionProvider (targeting Hexagon NPU via QnnHtp.dll in burst mode) as its primary provider. On non-Snapdragon host development machines without onnxruntime-qnn, Netra gracefully defaults to CPUExecutionProvider. This architectural separation ensures immediate target readiness for Snapdragon NPU acceleration without breaking cross-platform developer deployment.

6. Accessibility & Real-World Use Cases

- **Medication & Product Safety:** Point camera at pill bottles or food items; Netra detects the bottle and reads the dosage/labels out loud.
- **Navigation & Sign Reading:** Assists visually impaired individuals in recognizing door signs, room numbers, or printed directions.
- **Item Identification:** Helps low-vision users locate household items (keys, cups, books, electronics) hands-free.
- **Privacy-First Assistance:** Operates 100% offline with zero video or image telemetry leaving the device.

7. Key Innovations

- **Zero-Cloud Pipeline:** Complete visual perception, OCR, and TTS run locally without network requests.
- **Ultra-Compact Storage Footprint:** Total model binaries require less than 15 MB of disk space.
- **Multi-Stage Perceptual Fusion:** Synchronizes visual bounding boxes with recognized text into a single coherent spoken summary.

8. Deployment & Operation

Netra runs directly on Windows 11 PCs. Users launch the application via python main.py or double-clicking run.bat. The application automatically initializes camera capture, loads local ONNX models into memory, and displays the high-contrast dashboard window. Pressing the SPACE bar triggers a complete vision-to-speech turn.

9. Verified Scope & Technical Limitations

- **NPU Hardware Benchmark:** QNN Execution Provider initialization code is complete and verified, but physical NPU benchmarks require deployment on a Snapdragon ARM64 Copilot+ device.
- **OCR Latency:** EasyOCR CPU execution requires ~1.5 seconds per turn on host processor.
- **Illumination Sensitivity:** Text recognition quality is subject to environmental lighting and camera focus clarity.

10. Future Roadmap

- **Full QNN NPU Graph Compilation:** Exporting OCR and language models directly to Qualcomm AI Hub QNN graph format.
- **Voice-Activated Queries:** Integrating offline local keyword spotting for hands-free audio prompting.
- **Spatial Depth Assistance:** Integrating monocular depth estimation to speak object distances.